Name: _____

Newton's Laws of Motion Activities

1st Law of Motion

What is Newton's first law of motion?

Activity 1: Lazy Coins

You will need: A piece of A4 paper, five 20c coins

What to do:

1. Place the paper on a desk so that about 5cm of the short side hangs over the edge.
2. Place a 20c coin on the opposite end of the paper.
3. Hold the end of the paper hanging over the edge of the desk and pull it very quickly. If you do this fast enough, the coin should remain unmoved.
4. Repeat the steps above, but with two coins stacked on top of each other. Keep going to see how many coins you can do this with.
5. Repeat this activity, but pulling the paper slowly rather than quickly.

Questions:

1. What force acted between the paper and the coins as the paper slid beneath the coins?

2. Why can you leave the coins unmoved when you pull the paper out quickly?

3. Why could you not leave the coins unmoved when the paper was pulled out slowly?

Activity 2: Tricky Tricks

You will need: A clothes peg, a small square of cardboard, a 10c coin.

What to do:

1. Stand the peg on one end.
2. Place the cardboard on top of the peg.
3. Place the coin on top of the cardboard.
4. Holding the peg at the base, try to flick the cardboard out from under the coin, keeping the coin balanced on top of the peg.
5. For a challenge, try to do this without holding the peg still.

Questions:

1. Explain why the coin can remain on top of the peg using Newton's first law.

Activity 3: Crazy Coins

You will need: Five coins, all the same size

What to do:

1. Place one coin on the desk.
2. Put another coin about 10 cm in front of the first coin.
3. Flick the second coin into the first. Describe what happens.
4. Stack two coins on top of each other.
5. Place another coin 10 cm in front of the stack.
6. Flick the coin into the stack. Describe what happens.
7. Repeat with a stack of three coins and four coins.

Questions:

1. What did you observe about the coins moved by the flicked coin?

2. Explain your observations in terms of Newton's 1st Law.

As a challenge, try flicking a stack of two coins into a stack of three or four.

Activity 4: Crash Test Dummies

You will need: A toy car, plasticine, two books (like a dictionary), a piece of cardboard, baby powder

What to do:

1. Use the cardboard and a book to make a slope.
2. Make a small person from the plasticine.
3. Dust your person with baby powder so it loses some of its stickiness.
4. Sit your person on top of the car.
5. Gently push the car down the slope.
6. Note where the car comes to a stop.
7. Put the other book between the bottom of the ramp and where the car stops.
8. Roll the car down the slope again.

Questions:

1. What happened to the plasticine person when the car hit the book?

2. Explain your observations in terms of Newton's 1st Law.

2nd Law of Motion

What is Newton's second law of motion?

Activity 1: Creating Craters

You will need: Sand, large flat container, two rocks of different sizes

What to do:

1. Fill the container with sand and smooth the surface.
2. Drop the two rocks from the same height into the sand.
3. Carefully remove the rocks and compare the diameter of the craters created.

Questions:

1. Which rock made the bigger crater?

2. Explain your observations in terms of Newton's second law.

Activity 2: Newton's Race

You will need: Toy cars, plastic cup, sticky tape, wooden board, piece of cardboard, six 50g masses, masking tape, two bricks, coloured dots, a metre ruler

What to do:

1. Stick the plastic cup to the roof of the toy car with sticky tape.
2. Stack the bricks on the end of a row of desks and make a slope with the wooden board.
3. Put the cardboard on top of the wooden board and tape the end to the desk.
4. Run your car down the slope three times, placing a sticky dot where it stops each time.
5. Measure the distances from the base of the slope to each of the dots.
6. Record your measurements in the table below.

7. Add two masses to your cup, and then run the car down the slope three more times.
8. Measure and record.
9. Repeat the process with 4 and 6 masses.

Results:

Questions:

1. How does increasing the mass affect the force (in this case the distance the car rolls) of objects in motion? Explain your answer using the data you have collected.

2. What would happen if you added 8 masses to the car? Predict how far it would roll.

3. Explain the results of your experiment in terms of Newton's second law.

Activity 3: Mass vs Acceleration

You will need: Motion trolley, set of 10 slotted masses, pulley and clamp, string, 3 bricks, stopwatch, ruler

What to do:

1. Clamp the pulley onto the edge of a desk.
2. Tie one end of the string onto the trolley.
3. Tie the other end of the string onto the slotted masses.
4. Thread the string over the pulley.
5. Pull the trolley back and release it to check it runs freely.

Mass Added	Distance 1	Distance 2	Distance 3	Average
0 g				
100 g				
200 g				
300 g				

6. Mark a spot on the desk where you will release the trolley from each time.
7. Measure the distance from this spot to the pulley.
8. Pull the trolley back to the marked spot.
9. Release the trolley and measure the time it takes to reach the pulley. Record the time below.
10. Repeat twice more and calculate an average.
11. Add one brick to the trolley and repeat the process.
12. Continue until you have results for two and three bricks.

Results:

Distance: _____

Number of Bricks	Time of Trial 1	Time of Trial 2	Time of Trial 3	Average Time	Acceleration
0					
1					
2					
3					

Questions:

1. Calculate the acceleration of the trolley for each mass using the formula

$$a = \frac{2d}{t^2}$$ where d = the distance travelled

t = the average time

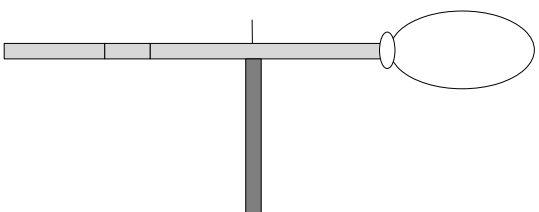
2. What happened to the acceleration of the trolley as you increased the mass?

3. Explain your observation in terms of Newton's second law.

3rd Law of Motion

What is Newton's third law of motion?

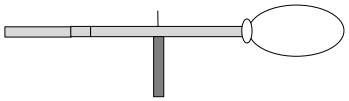
Activity 1: Fun with Balloons



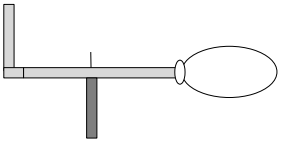
You will need: Balloon, flexible straw, sticky tape, pencil with an eraser on the end, pin

What to do:

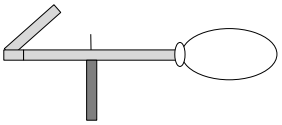
1. Attach the balloon to the end of the straw which is furthest away from the bend.
2. Push the pin straight through the straw about halfway between the balloon and the bend in the straw.
3. Push the end of the pin into the eraser on the pencil.
4. Blow up the balloon, but leave the straw straight. Release the air in the balloon. What happens?



5. Blow up the balloon and bend the straw to a 90° angle before allowing the air to escape. What happens?



6. Blow up the balloon again and bend the straw to a 45° angle before allowing the air to escape. What happens?



7. Remove the pin from the straw and blow up the balloon. Release the straw. What happens?

Questions:

1. What did you notice about the direction of the air flow and the direction of the balloon?

2. Explain your observation in terms of Newton's third law.

Activity 2: Film Canister Rocket

You will need: Film canister, water, alka-seltzer tablet

What to do:

1. Half fill the film canister with water.
2. Break an alka-seltzer tablet in half.
3. Drop half of the alka-seltzer tablet into the water in the film canister.
4. Place the lid firmly on the film canister.
5. Turn it upside down so that it is sitting on the lid of the film canister and place in an open area.

Questions

1. Describe what happens to the film canister.

2. Draw an action-reaction diagram to describe what is happening in your canister.

3. Explain what is happening using Newton’s third law.

4. Describe a real life example where you have seen Newton’s third law in action.